Islamic University of Technology

Lab 3: Formatted IO

CSE 4108 : Structured Programming Lab

Section 1B

MIA, NFS, IAC, ASH

Sep 17, 2024

1 Lab Tasks

This lab is dedicated for formatted IO practice. Make sure that the task outputs match exactly with the sample outputs given in the question.

1.1 It's a Date

Write a C program that uses `scanf` function to scan a date in the format *dd-mm-yyyy* (e.g. 07-07-2021) and print it in *yyyy/mm/dd* format.

[Sample Program]
Input

```
Enter date (dd-mm-yyyy): 05-12-2020
```

Output

```
Formatted date (yyyy/mm/dd): 2020/12/05
```

1.2 Shocktail

Natasha Raven has started her own fruit juice bar called 'Tropical Relief'. Recently, she has come up with a new unique drink with a mixture of 3 fruit juice cocktail, which she is calling the "Shocktail". The drink contains mango, orange and apple juice and the amounts are highly configurable by the customers. Each drink can contain from 0.0 up to 100.00 milliliters of each juice.

Write a C program that takes input the amount of each juice in milliliters and prints out a receipt that contains the amount and percentage of each individual juices along with the total amount in milliliters.

[Sample Program]
Input

```
Enter mango juice amount (ml): 10.532
Enter orange juice amount (ml): 14.2234
Enter apple juice amount (ml): 5.5
```

Output

```
Shocktail Receipt - Tropical Relief
Mango Juice:     10.53ml         34.81%
Orange Juice:    14.22ml         47.01%
Apple Juice:      5.50ml         18.18%
Total Juice:     30.26ml        100.00%
```

[Hint] *Use tab ('\t') to align texts.*

1.3 Telephony Business

In Bangladesh, phone numbers provided by mobile operators are formatted as, `+880 XX NNNNNNNN`, where `+880` is the country code, `XX` is the operator and `NNNNNNNN` is the serial number.

Write a C program that would take a phone number as input in the format shown in the sample program and output the different components of the number.

[Sample Program]
Input

```
Enter phone number: +880 17 12345678
```

Output

```
Country Code:   +880
Operator Code:  17
Serial Number:  12345678
```

[Hint] *Use tab ('\t') to align texts.*

1.4 Haxxor's Symbol

Revi Hackerman is a famous hacker. Despite the numerous things he has hacked, he still feels like his tasks are incomplete for the lack of a signature. Recently he has come up with a plan and that is to leave his signature in the console of every hacked computer after he is done. He decided to only use `printf` function and `return 0` statements inside the main function in his C program with some additional rules.

Help Revi by Writing a C program that would print the given symbol on console. However, the **print statements can contain only two conversion specifications within the format string with a newline escape sequence and nothing else** e.g `printf("%(-)m.pX%(-)m.pX\n", exp1, exp2);`. Use of multiple `printf` statements are allowed. Use of `return 0` statement is also allowed.

[Symbol]

```
1         1
1         1
1         1
0000000000
1         1
1         1
1         1
```

[Hint] *Use integer conversion specifications along with appropriate modifications and justification and pass 0s or 1s as expressions of* `printf` *functions.*

1.5 Fractional Addition

Write a C program that would take two fractions with plus(+) or minus (-) sign in the middle and show their addition as another fraction. There is no need to represent the output fraction in its reduced form.
[Sample Program]
Input

```
Enter fractions: 5/6+3/4
```

Output

```
Sum: 38/24
```

[Hint] *C code to enter a scan a single fraction is* `scanf("%d /%d", &nume, &denom);`. *The plus or minus sign in the middle can be considered as part of the numerator of the second fraction. The fractional sum can be calculated using equation 1.*

$$\frac{a}{b} + \frac{x}{y} = \frac{(a \times y) + (b \times x)}{b \times y} \tag{1}$$

1.6 Base Conversion

Given an input of an integer number, show it as an octal and hexadecimal number.

[Sample Program]

Input

```
Enter an integer number: 45
```

Output

```
45 in Hexadecimal: 2D
45 in Octal: 55
```

[Hint] *Use appropriate format specifiers in printing.*